Capital-Level Factor Neutrality with Projected Partial Rebalancing

Anonymous Author(s)

Abstract

Active managers often remove factor exposure from alpha signals before portfolio construction. This signal-level operation is useful for interpretation, but it does not generally imply that the implemented portfolio is factor-neutral after risk scaling, clipping, gross normalisation, volatility targeting, or partial rebalancing. This paper studies a portfolio-construction method that imposes neutrality directly on the capital weights that are actually traded.

At each rebalance, an exposure matrix F_t defines the monitored neutral subspace $\ker F_t^\top$. The portfolio is written as $u_t = Q_t\theta_t$, where the columns of Q_t form an orthonormal basis of this subspace. Optimisation is therefore carried out in neutral coordinates, so every feasible portfolio satisfies the factor-neutrality equations on the implemented capital vector. The same representation gives a direct equality-constrained Markowitz interpretation and finite-sample exposure leakage bounds when the construction and monitoring exposure matrices differ.

The paper also introduces a projected partial-rebalancing rule. Before trading toward the new neutral target, the previous holding is projected into the current neutral subspace; the portfolio is then moved part way toward the full neutral target. Since both endpoints are neutral under the current exposure matrix, the entire interpolation path remains neutral.

Walk-forward tests across six research universes compare ordinary alpha, residualised alpha, the full neutral target, and projected partial neutral-coordinate portfolios. The results are used as construction diagnostics: neutral-coordinate portfolios exhibit numerical-level exposure leakage, while projected partial variants reduce turnover relative to the full neutral target. The empirical section evaluates exposure control, turnover, and implementation stability under fixed validation choices.

Keywords: portfolio construction; factor neutrality; turnover control; constrained Markowitz optimisation; transaction costs; walk-forward validation.

1 Introduction

Active portfolio management separates forecasting from construction. A forecasting model produces scores, expected-return proxies, or rankings; the construction process converts those inputs into capital weights subject to risk, leverage, turnover, benchmark, liquidity, and factor-exposure controls. A common practice is to residualise the alpha signal against style, sector, beta, volatility, or other characteristics before optimisation. That operation can clarify the signal, but it does not control the exposures of the implemented capital vector. Risk weighting, clipping, gross normalisation, volatility targeting, and partial rebalancing can all reintroduce factor exposure.

For an active sleeve, overlay, or monitored mandate, the relevant question is whether the traded capital vector satisfies the exposure equations after all construction steps have been applied. The problem is finite-sample and dynamic: the investable universe, exposure matrix, risk estimate, and neutral subspace change through time.

At rebalance time t, let $F_t \in \mathbb{R}^{n_t \times m_t}$ be the chosen exposure matrix. Its columns may contain sector indicators, style scores, beta estimates, volatility characteristics, nonlinear transformations of characteristics, or any other exposure directions to be removed. A capital vector $u_t \in \mathbb{R}^{n_t}$ is neutral when

$$F_t^\top u_t = 0.$$

In words, the dollar-weighted exposure of the implemented active portfolio to each selected exposure column is zero. Rather than residualising the signal and then optimising in the original asset coordinates, we compute an orthonormal basis Q_t of $\ker F_t^\top$ and optimise over

$$u_t = Q_t \theta_t.$$

Thus the optimisation variables θ_t are coordinates inside the neutral capital space; any feasible coordinate vector automatically produces a neutral capital vector. This neutral-coordinate representation distinguishes capital-space neutrality from signal-space residualisation: the feasible capital vectors are neutral, not just the forecasting input.

Turnover is the second issue. A fully reoptimised neutral target can move sharply when signals, covariances, or exposure definitions change. Projected partial neutral-coordinate rebalancing first projects the old holding into the current neutral subspace and then chooses a point on the line segment to the full neutral target. A one-dimensional utility calculation selects the interpolation coefficient. Since both endpoints lie in the current neutral subspace, the interpolation remains neutral.

The contribution is a finite-sample construction protocol rather than an alpha model or covariance estimator. The null-space representation itself is a standard linear-algebraic device. Its role here is to make neutrality a property of the implemented capital vector after optimisation, normalisation, overlays, and partial-trading rules have entered the process. This distinction matters in practice because residualised signals can still lead to non-neutral portfolios once these steps are applied.

The paper makes three specific points. First, capital-level neutrality should be imposed and monitored on the final portfolio weights, not inferred from a residualised forecasting signal. Second, neutral coordinates give a simple finite-sample implementation of equality-constrained portfolio construction and make ex-post exposure checks transparent. Third, when the exposure matrix changes through time, partial rebalancing should begin by projecting the previous holding into the current neutral subspace; otherwise a turnover-saving rule can carry stale exposure into the new period.

The empirical analysis tests these construction claims using walk-forward research panels. The tests ask whether the proposed construction enforces the monitored exposure constraints after optimisation and rebalancing, and whether partial rebalancing can reduce turnover while preserving capital-level neutrality. The fixed validation specification uses a cubic volatility-exposure matrix, while the theory applies to any user-chosen finite exposure matrix. The empirical role of the tests is to validate the construction layer rather than to rank alpha models.

2 Related literature and positioning

The paper is related to mean–variance portfolio construction, constrained optimisation, and active factor control. The starting point is Markowitz (1952). Subsequent work shows that estimation error, constraints, and implementation rules can materially affect realised portfolios; see, for example, Michaud (1989), DeMiguel et al. (2009), and Kolm et al. (2014). Covariance shrinkage and portfolio constraints are often used to stabilise realised allocations (Ledoit and Wolf, 2004; Jagannathan and Ma, 2003). Transaction costs and rebalancing rules are also central in practical portfolio construction (Lobo et al., 2007; Boyd et al., 2017).

The paper is also related to active portfolio management and factor-neutral mandates. In practice, alpha signals may be residualised against sector, style, beta, volatility, liquidity, or other exposure variables before optimisation; see Grinold and Kahn (2000) and Clarke et al. (2002). Such residualisation is useful for signal interpretation and for diagnosing whether a forecast is mainly a proxy for known characteristics. It is not, however, the same as constraining the traded capital vector. A risk model, leverage rule, gross normalisation, volatility overlay, clipping rule, or partial-rebalancing rule can change the exposures of the implemented book.

The method is a portfolio-construction module. It does not replace the forecasting model, covariance estimator, transaction-cost model, or risk system. Given an exposure matrix, it enforces neutrality on the portfolio weights produced by the construction process. The point is not the existence of equality-constrained optimisers, but where the constraint is imposed and how stale exposure is handled under partial rebalancing.

3 Asset-management workflow and use case

The method is intended as a factor-neutral construction module. The manager first chooses the exposure matrix F_t to be neutralised. Its columns may include sector indicators, style scores, beta estimates, volatility characteristics, liquidity variables, or nonlinear transformations of such variables. The neutral basis Q_t is then computed using information available before the holding period.

For an asset manager, the relevant audit object is the implemented portfolio, not the raw forecast. A signal may be orthogonal to the monitored exposures at the forecasting stage and still produce a book with nonzero exposure after risk weighting, gross normalisation, volatility targeting, or partial trading. The proposed workflow treats factor neutrality as a post-construction requirement: the exposure check is applied to the capital vector that would actually be traded.

The intended use case is an active sleeve or overlay in which neutrality is monitored after optimisation and rebalancing. The alpha vector and risk estimate are projected into the neutral capital space to obtain a full neutral target. The previous holding is separately projected into the current neutral subspace, and the final portfolio moves part way from the projected old holding to the new full target. The monitor records exposure leakage, turnover, the selected rebalancing coefficient, ex-ante risk, conditioning, drawdown, numerical rank, and implementation flags at each rebalance.

This separates three implementation choices: the forecasting signal, the exposure policy, and the speed of rebalancing. In an existing portfolio-construction system, the method can be inserted after signal generation and before trade generation: choose the monitored exposure matrix, compute the neutral target, project the old holding into the current neutral subspace, select the rebalancing coefficient, and monitor realised exposure leakage and turnover. The guarantees apply to the exposure columns included in F_t.

Projected partial neutral-coordinate rebalancing.
At rebalance time t:

1. Choose the monitored exposure matrix F_t using information available before the holding period.
2. Compute an orthonormal basis Q_t for $\ker F_t^\top$.
3. Construct the full neutral target $u_t^* = Q_t\theta_t^*$.
4. Project the previous holding into the current neutral subspace:
$$\widetilde{u}_{t-1} = Q_t Q_t^\top u_{t-1}.$$
5. Trade only part way toward the full neutral target:
$$u_t(\beta) = (1-\beta)\widetilde{u}_{t-1} + \beta u_t^*, \qquad 0 \le \beta \le 1.$$
6. Choose β from the one-dimensional objective that trades off expected return, risk, transaction costs, and turnover.

Because both $\widetilde{u}_{t-1}$ and u_t^* lie in $\ker F_t^\top$, every portfolio $u_t(\beta)$ is neutral under the current exposure matrix.

4 Neutral-coordinate theory

4.1 Exact finite-sample nonlinear neutrality

Let n_t denote the number of assets available at time t and let m_t denote the number of exposure functions to be neutralised. In the implementation tested below, $F_t \in \mathbb{R}^{n_t \times m_t}$ is built from nonlinear functions of a characteristic. For example, if $\sigma_{i,t}$ is an asset-level volatility characteristic, one may take

$$F_t = \begin{bmatrix} 1 & \sigma_{1,t} & \sigma_{1,t}^2 & \sigma_{1,t}^3 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \sigma_{n_t,t} & \sigma_{n_t,t}^2 & \sigma_{n_t,t}^3 \end{bmatrix}. \tag{1}$$

The empirical implementation uses the cubic specification $F_t = [1, \sigma_t, \sigma_t^2, \sigma_t^3]$ as a parsimonious way to remove level, linear, quadratic, and cubic exposure to the volatility characteristic while retaining sufficient degrees of freedom in the smaller universes. Here $\sigma_{i,t}$ denotes the 252-day trailing volatility of asset i, computed using only information available before the holding interval. Thus neutrality with respect to F_t removes not only linear exposure to the characteristic, but also the specified nonlinear exposure directions. The construction is not tied to this particular choice: sector dummies, style factors, beta estimates, liquidity characteristics, or other finite-sample exposure functions can be included as columns of F_t. Section 7.3 reports order-sweep checks.

Let
$$\mathcal{N}_t := \ker F_t^\top,$$
and choose an orthonormal basis $Q_t \in \mathbb{R}^{n_t \times d_t}$ of $\mathcal{N}_t$, where $d_t = n_t - \operatorname{rank}(F_t)$. A neutral-coordinate portfolio is a capital vector of the form

$$u_t = Q_t\theta_t, \qquad \theta_t \in \mathbb{R}^{d_t}. \tag{2}$$

Thus θ_t is not a second portfolio; it is the coordinate vector of the implemented capital weights inside the neutral subspace. Throughout the numerical statement below, $\varepsilon_{\mathrm{svd}}$ denotes an absolute singular-value threshold. If a relative tolerance is used in code, it should be multiplied by the leading singular value before applying the bound.

Theorem 1 (Exact finite-sample nonlinear neutrality)**.** *Every neutral-coordinate portfolio $u_t = Q_t\theta_t$ satisfies*

$$F_t^\top u_t = 0. \tag{3}$$

Consequently, the portfolio is exactly neutral to every exposure column included in F_t, including non-linear characteristic columns such as σ_t^2 and σ_t^3.

Moreover, suppose Q_t is computed from a singular value decomposition of F_t. If the retained numerical null-space vectors correspond to singular values at most $\varepsilon_{\mathrm{svd}}$, then

$$\|F_t^\top Q_t\|_{2\to 2} \le \varepsilon_{\mathrm{svd}}. \tag{4}$$

Hence the realised exposure leakage of any computed neutral-coordinate portfolio obeys

$$\|F_t^\top Q_t\theta_t\|_2 \le \varepsilon_{\mathrm{svd}}\|\theta_t\|_2. \tag{5}$$

If Q_t is an exact basis of $\ker F_t^\top$, then $\varepsilon_{\mathrm{svd}} = 0$ and the neutrality is exact.

Proof. Since the columns of Q_t lie in $\ker F_t^\top$, one has $F_t^\top Q_t = 0$. Therefore, for every θ_t, $F_t^\top u_t = F_t^\top Q_t\theta_t = 0$.

For the numerical statement, write a singular value decomposition

$$F_t = U_t\Sigma_t^F V_t^\top.$$

The left singular vectors u_j satisfy $F_t^\top u_j = \sigma_j v_j$. If Q_t is formed from left singular vectors whose singular values satisfy $\sigma_j \le \varepsilon_{\mathrm{svd}}$, then for every unit vector x,

$$\|F_t^\top Q_t x\|_2^2 = \sum_j \sigma_j^2 x_j^2 \le \varepsilon_{\mathrm{svd}}^2 \sum_j x_j^2 = \varepsilon_{\mathrm{svd}}^2.$$

Thus $\|F_t^\top Q_t\|_{2\to 2} \le \varepsilon_{\mathrm{svd}}$. The portfolio leakage bound follows by applying this operator bound to θ_t. □

This result is purely finite-sample. If the columns of Q_t lie in $\ker F_t^\top$, then the implemented weight vector has zero exposure to every column of F_t. The statement does not depend on a factor model being true; it depends only on the exposure matrix used at the rebalance.

4.2 Constrained Markowitz interpretation

Let a_t be the alpha vector, let Σ_t be the covariance or risk matrix used in construction, let $\gamma > 0$ be a risk-aversion parameter, and let $\mathcal{C} \subseteq \mathbb{R}^{n_t}$ denote the portfolio constraint set. The ordinary active optimiser solves

$$u_t^{\mathrm{ord}} = \arg\max_{u\in\mathcal{C}} \left\{ a_t^\top u - \frac{\gamma}{2} u^\top \Sigma_t u \right\}. \tag{6}$$

The neutral-coordinate optimiser imposes neutrality by construction. Define

$$\Theta_t := \{\theta \in \mathbb{R}^{d_t} : Q_t\theta \in \mathcal{C}\}. \tag{7}$$

Then the constrained neutral-coordinate problem is

$$\theta_t^* = \arg\max_{\theta\in\Theta_t} \left\{ a_t^\top Q_t\theta - \frac{\gamma}{2}\theta^\top Q_t^\top \Sigma_t Q_t\theta \right\}, \qquad u_t^{\mathrm{NC}} = Q_t\theta_t^*. \tag{8}$$

Proposition 1 (Neutral constrained optimality)**.** *The neutral-coordinate optimiser* u_t^{NC} *is exactly the Markowitz optimiser restricted to the neutral feasible set:*

$$u_t^{\mathrm{NC}} = \underset{u \in \mathcal{C} \cap \ker F_t^\top}{\arg\max} \left\{ a_t^\top u - \frac{\gamma}{2} u^\top \Sigma_t u \right\}. \tag{9}$$

Proof. Because the columns of Q_t form an orthonormal basis of $\mathcal{N}_t = \ker F_t^\top$, the map $\theta \mapsto Q_t\theta$ is a one-to-one parametrisation of $\mathcal{N}_t$. For every $\theta \in \mathbb{R}^{d_t}$, $Q_t\theta \in \mathcal{N}_t$, and for every $u \in \mathcal{N}_t$, one has $u = Q_t Q_t^\top u$. Thus every neutral portfolio has a unique coordinate vector $\theta = Q_t^\top u$. The additional constraint $u \in \mathcal{C}$ is exactly the condition $\theta \in \Theta_t$. Therefore optimising over $\theta \in \Theta_t$ is equivalent to optimising over $u \in \mathcal{C} \cap \ker F_t^\top$. Substituting $u = Q_t\theta$ gives the stated objective. □

Corollary 1 (Basis invariance of neutral coordinates)**.** *Let* $Q_t, \widetilde{Q}_t \in \mathbb{R}^{n_t \times d_t}$ *be two orthonormal bases of*

$$\mathcal{N}_t := \ker F_t^\top.$$

Then

$$\{Q_t\theta : \theta \in \mathbb{R}^{d_t}\} = \{\widetilde{Q}_t\eta : \eta \in \mathbb{R}^{d_t}\} = \mathcal{N}_t.$$

Consequently, in exact arithmetic, the neutral-coordinate feasible capital set is independent of the particular orthonormal null-space basis used to represent it. If the constrained objective has a unique maximiser on $\mathcal{C} \cap \mathcal{N}_t$*, then the resulting capital vector* u_t^* *is basis-invariant. Different choices of* Q_t *may change the coordinate vector, but not the optimal portfolio in capital space.*

Proof. Since both Q_t and $\widetilde{Q}_t$ have columns forming orthonormal bases of $\mathcal{N}_t$, their column spaces are equal:

$$\mathrm{Range}(Q_t) = \mathrm{Range}(\widetilde{Q}_t) = \mathcal{N}_t.$$

Therefore every vector of the form $Q_t\theta$ lies in $\mathcal{N}_t$, and every vector in $\mathcal{N}_t$ can be represented as $Q_t\theta$ for a unique $\theta = Q_t^\top u$. The same statement holds for $\widetilde{Q}_t$. This proves equality of the feasible capital sets.

Now suppose the objective has a unique maximiser over $\mathcal{C} \cap \mathcal{N}_t$. Since the feasible set in capital space is the same for Q_t and $\widetilde{Q}_t$, both coordinate representations optimise the same objective over the same set $\mathcal{C} \cap \mathcal{N}_t$. Hence they produce the same unique capital vector u_t^*, although the coordinates are related by an orthogonal change of basis. □

Computationally, the SVD basis is merely a numerical coordinate chart on the neutral subspace. The economically meaningful object is the capital vector u_t, not the particular coordinate vector θ_t.

4.3 Closed-form neutral optimiser

The null-coordinate construction is equivalent to the classical equality-constrained Markowitz problem. This closed-form version clarifies what neutral coordinates do: they do not alter the objective, but choose coordinates in which the neutrality constraint is automatic. For this subsection, ignore box, leverage, and gross-exposure constraints and consider only the finite-sample neutrality constraint:

$$\max_{u \in \mathbb{R}^{n_t}} \left\{ a_t^\top u - \frac{\gamma}{2} u^\top \Sigma_t u \right\} \quad \text{subject to} \quad F_t^\top u = 0. \tag{10}$$

Proposition 2 (Closed-form neutral Markowitz solution)**.** *Assume* $\Sigma_t \succ 0$ *and* $F_t^\top \Sigma_t^{-1} F_t$ *is nonsingular. Then the equality-constrained optimiser is*

$$u_t^{\mathrm{neu}} = \frac{1}{\gamma} \left[\Sigma_t^{-1} a_t - \Sigma_t^{-1} F_t (F_t^\top \Sigma_t^{-1} F_t)^{-1} F_t^\top \Sigma_t^{-1} a_t \right]. \tag{11}$$

Equivalently,

$$u_t^{\mathrm{neu}} = \frac{1}{\gamma} P_t^{\Sigma} \Sigma_t^{-1} a_t, \tag{12}$$

where

$$P_t^{\Sigma} := I - \Sigma_t^{-1} F_t (F_t^\top \Sigma_t^{-1} F_t)^{-1} F_t^\top \tag{13}$$

is the Σ_t*-metric projection onto* $\ker F_t^\top$*. If* Q_t *is an orthonormal basis of* $\ker F_t^\top$*, the same optimiser is obtained in neutral coordinates:*

$$\theta_t^* = \frac{1}{\gamma} (Q_t^\top \Sigma_t Q_t)^{-1} Q_t^\top a_t, \qquad u_t^{\mathrm{neu}} = Q_t \theta_t^*. \tag{14}$$

This formula is the usual unconstrained Markowitz allocation with the component that would create exposure to F_t *removed in the* Σ_t*-metric. The neutral-coordinate formula gives the same portfolio by optimising directly inside the neutral subspace.*

Proof. The Lagrangian is

$$\mathcal{L}(u, \lambda) = a_t^\top u - \frac{\gamma}{2} u^\top \Sigma_t u - \lambda^\top F_t^\top u.$$

The first-order condition with respect to u is

$$a_t - \gamma \Sigma_t u - F_t \lambda = 0.$$

Hence

$$u = \frac{1}{\gamma} \Sigma_t^{-1} (a_t - F_t \lambda).$$

Imposing $F_t^\top u = 0$ gives

$$F_t^\top \Sigma_t^{-1} F_t \, \lambda = F_t^\top \Sigma_t^{-1} a_t.$$

Since $F_t^\top \Sigma_t^{-1} F_t$ is nonsingular,

$$\lambda = (F_t^\top \Sigma_t^{-1} F_t)^{-1} F_t^\top \Sigma_t^{-1} a_t.$$

Substituting this expression into the formula for u yields the displayed closed form.

For the coordinate form, write $u = Q_t \theta$. The objective becomes

$$a_t^\top Q_t \theta - \frac{\gamma}{2} \theta^\top Q_t^\top \Sigma_t Q_t \theta.$$

Since $\Sigma_t \succ 0$ and Q_t has full column rank, $Q_t^\top \Sigma_t Q_t \succ 0$. The first-order condition is

$$Q_t^\top a_t - \gamma Q_t^\top \Sigma_t Q_t \theta = 0,$$

hence

$$\theta_t^* = \frac{1}{\gamma} (Q_t^\top \Sigma_t Q_t)^{-1} Q_t^\top a_t.$$

Because $Q_t \theta$ parametrises exactly $\ker F_t^\top$, this coordinate solution is the same constrained optimiser. □

Remark 1. *If* $F_t^\top \Sigma_t^{-1} F_t$ *is singular, the inverse in Proposition 2 may be replaced by the Moore–Penrose pseudoinverse. The neutral-coordinate formulation avoids this particular saddle-system inversion by working directly in a basis of the neutral subspace.*

4.4 Relation to signal residualisation

A common alternative to neutral portfolio construction is to residualise the alpha signal against the exposure matrix. Neutral coordinates are different: they impose neutrality on the final capital allocation. The two operations coincide only under special risk geometry. Let

$$P_t := Q_t Q_t^\top$$

be the Euclidean projector onto $\mathcal{N}_t = \ker F_t^\top$. The Euclidean residualised alpha is

$$a_t^{\mathrm{res}} := P_t a_t.$$

Proposition 3 (Residualisation agrees with neutral-coordinate only in special cases)**.** *Assume there are no box, leverage, or gross-exposure constraints.*

1. *If $\Sigma_t = \sigma_t^2 I$, then the neutral-coordinate optimiser is*

$$u_t^{\mathrm{NC}} = \frac{1}{\gamma \sigma_t^2} P_t a_t.$$

   Thus, under isotropic risk, neutral-coordinate optimisation is proportional to the residualised alpha signal.

2. *For general $\Sigma_t \succ 0$, the neutral-coordinate optimiser is*

$$u_t^{\mathrm{NC}} = \frac{1}{\gamma} Q_t (Q_t^\top \Sigma_t Q_t)^{-1} Q_t^\top a_t.$$

   In general this is not proportional to $P_t a_t$. Therefore, residualising the alpha signal is not equivalent to imposing neutrality on the final capital allocation.

Proof. If $\Sigma_t = \sigma_t^2 I$, then

$$Q_t^\top \Sigma_t Q_t = \sigma_t^2 Q_t^\top Q_t = \sigma_t^2 I.$$

The coordinate solution from Proposition 2 therefore gives

$$u_t^{\mathrm{NC}} = Q_t \left[\frac{1}{\gamma \sigma_t^2} Q_t^\top a_t \right] = \frac{1}{\gamma \sigma_t^2} Q_t Q_t^\top a_t = \frac{1}{\gamma \sigma_t^2} P_t a_t.$$

For general Σ_t, the restricted risk matrix $Q_t^\top \Sigma_t Q_t$ need not be a scalar multiple of the identity. The optimiser depends on the inverse of this restricted risk matrix, which generally differs from the Euclidean projection $Q_t Q_t^\top a_t$. The two agree only when the risk matrix acts isotropically on the neutral subspace, or in other special cases where the restricted inverse reduces to a scalar multiple of the identity. □

Proposition 4 (Residualised alpha need not produce neutral capital)**.** *Let $a_t^{\mathrm{res}} = P_t a_t$, and consider the unconstrained optimiser using the residualised alpha:*

$$u_t^{\mathrm{res}} = \frac{1}{\gamma} \Sigma_t^{-1} a_t^{\mathrm{res}} = \frac{1}{\gamma} \Sigma_t^{-1} P_t a_t. \tag{15}$$

Then

$$F_t^\top u_t^{\mathrm{res}} = \frac{1}{\gamma} F_t^\top \Sigma_t^{-1} P_t a_t. \tag{16}$$

Thus residualising the signal guarantees neutral capital for all a_t if and only if

$$F_t^\top \Sigma_t^{-1} P_t = 0. \tag{17}$$

Equivalently, Σ_t^{-1} must map the neutral subspace into itself. This condition is not satisfied by a general risk matrix.

Proof. The displayed expression follows by direct substitution:

$$F_t^\top u_t^{\mathrm{res}} = F_t^\top \left[\frac{1}{\gamma}\Sigma_t^{-1} P_t a_t\right] = \frac{1}{\gamma} F_t^\top \Sigma_t^{-1} P_t a_t.$$

This is zero for every alpha vector a_t if and only if the matrix $F_t^\top \Sigma_t^{-1} P_t$ is identically zero. Since P_t projects onto $\mathcal{N}_t = \ker F_t^\top$, this condition says precisely that applying Σ_t^{-1} to a neutral vector remains neutral. A general positive definite risk matrix does not preserve this subspace, so residualising alpha is generally weaker than imposing neutrality on the capital vector itself. □

Thus residualising the alpha vector and constraining the capital vector are different operations. They agree in special cases, for example under isotropic risk, but not for a general covariance matrix. This is why the empirical comparison includes both a residualised-alpha baseline and neutral-coordinate portfolios.

4.5 Robustness to misspecified exposures

The exact neutrality theorem is finite-sample and model-relative: it enforces neutrality with respect to the exposure matrix F_t used in the construction. If the realised or target exposure matrix differs from F_t, the remaining leakage is controlled by the exposure misspecification. Let

$$\widetilde{F}_t = F_t + \Delta_t$$

be a perturbed exposure matrix. Since $F_t^\top u_t^{\mathrm{NC}} = 0$,

$$\widetilde{F}_t^\top u_t^{\mathrm{NC}} = (F_t + \Delta_t)^\top u_t^{\mathrm{NC}} = \Delta_t^\top u_t^{\mathrm{NC}}.$$

Proposition 5 (Exposure-misspecification leakage bound)**.** *Let $u_t^{\mathrm{NC}} \in \ker F_t^\top$, and let $\widetilde{F}_t = F_t + \Delta_t$. Then*

$$\|\widetilde{F}_t^\top u_t^{\mathrm{NC}}\|_2 \le \|\Delta_t\|_{2\to 2} \|u_t^{\mathrm{NC}}\|_2. \tag{18}$$

If additionally $\|u_t^{\mathrm{NC}}\|_1 \le G$, then

$$\|\widetilde{F}_t^\top u_t^{\mathrm{NC}}\|_2 \le G\|\Delta_t\|_{2\to 2}. \tag{19}$$

Proof. Since $F_t^\top u_t^{\mathrm{NC}} = 0$,

$$\widetilde{F}_t^\top u_t^{\mathrm{NC}} = \Delta_t^\top u_t^{\mathrm{NC}}.$$

The first inequality follows from the operator norm bound. The second follows from $\|u\|_2 \le \|u\|_1$. □

This is the theorem-level object tested by the exposure-stress experiment: perturbing the exposure matrix should produce leakage that is linear in the size of the perturbation and bounded by the displayed operator-norm estimate.

Proposition 6 (Combined numerical and exposure-misspecification leakage)**.** *Let $Q_t \in \mathbb{R}^{n_t \times d_t}$ be a computed neutral-coordinate basis satisfying*

$$\|F_t^\top Q_t\|_{2\to 2} \le \varepsilon_{\mathrm{svd}}.$$

Let the realised exposure matrix be $\widetilde{F}_t = F_t + \Delta_t$. For every computed neutral-coordinate portfolio $u_t = Q_t\theta_t$,

$$\|\widetilde{F}_t^\top u_t\|_2 \le \varepsilon_{\mathrm{svd}}\|\theta_t\|_2 + \|\Delta_t\|_{2\to 2}\|u_t\|_2. \tag{20}$$

If, in addition, the portfolio has gross exposure bounded by $\|u_t\|_1 \le G$, then

$$\|\widetilde{F}_t^\top u_t\|_2 \le \varepsilon_{\mathrm{svd}}\|\theta_t\|_2 + G\|\Delta_t\|_{2\to 2}. \tag{21}$$

Proof. Using $\widetilde{F}_t = F_t + \Delta_t$ and $u_t = Q_t\theta_t$,

$$\widetilde{F}_t^\top u_t = F_t^\top Q_t\theta_t + \Delta_t^\top u_t.$$

Taking Euclidean norms and applying the triangle inequality gives

$$\|\widetilde{F}_t^\top u_t\|_2 \le \|F_t^\top Q_t\theta_t\|_2 + \|\Delta_t^\top u_t\|_2.$$

The assumed computed null-space error gives

$$\|F_t^\top Q_t\theta_t\|_2 \le \|F_t^\top Q_t\|_{2\to 2}\|\theta_t\|_2 \le \varepsilon_{\text{svd}}\|\theta_t\|_2.$$

Also $\|\Delta_t^\top u_t\|_2 \le \|\Delta_t\|_{2\to 2}\|u_t\|_2$. Combining the inequalities gives the first bound. If $\|u_t\|_1 \le G$, then $\|u_t\|_2 \le \|u_t\|_1 \le G$, which yields the second bound. $\square$

Thus exposure leakage has two distinct sources: numerical null-space error and exposure-definition error. The former is controlled by the SVD tolerance and the latter by the realised discrepancy Δ_t between the construction exposures and the exposures audited after trading.

5 Projected partial neutral-coordinate rebalancing

5.1 Parallel transport and projected partial rebalancing

The neutral basis Q_t may change from one rebalance date to the next. A first attempt to stabilise coordinates is to align the bases by an orthogonal Procrustes transformation. Given bases Q_{t-1} and Q_t with the same retained dimension $d_t = d_{t-1}$, choose

$$R_t^* = \underset{R^\top R = I}{\arg\min}\, \|Q_tR - Q_{t-1}\|_F. \tag{22}$$

If

$$Q_t^\top Q_{t-1} = U_tS_tV_t^\top$$

is a singular value decomposition, then $R_t^* = U_tV_t^\top$. The aligned basis $\widetilde{Q}_t = Q_tR_t^*$ still spans the current neutral subspace because

$$F_t^\top \widetilde{Q}_t = F_t^\top Q_tR_t^* = 0.$$

If the null-space dimension changes, one first restricts to the common retained dimension or uses the corresponding rectangular partial-isometry Procrustes problem. Empirically, basis alignment improves coordinate stability, but turnover still requires a trade-size rule. The final candidates use the project-then-partially-rebalance rule below.

Let $P_{\mathcal{N}_t} = Q_tQ_t^\top$ be the Euclidean projector onto $\mathcal{N}_t$. First project yesterday's held portfolio into the current neutral subspace:

$$u_{t-1\to t} := P_{\mathcal{N}_t}u_{t-1}. \tag{23}$$

Then compute the full neutral target $u_t^* \in \mathcal{N}_t$ and trade only part of the way:

$$u_t(\beta) = (1-\beta)u_{t-1\to t} + \beta u_t^*, \qquad \beta \in [0,1]. \tag{24}$$

The coefficient β controls trading speed: $\beta = 0$ keeps the projected old holding, $\beta = 1$ moves fully to the new neutral target, and intermediate values trade part way while remaining neutral. The implemented rule chooses β by the one-dimensional utility problem

$$\max_{\beta\in[0,1]} \left\{ a_t^\top u_t(\beta) - \frac{\gamma}{2}u_t(\beta)^\top \Sigma_t u_t(\beta) - c\|u_t(\beta) - u_{t-1}\|_1 \right\}. \tag{25}$$

Theorem 2 (Neutrality and global solvability of projected partial neutral-coordinate rebalancing)**.** *Let Q_t be an orthonormal basis of $\mathcal{N}_t = \ker F_t^\top$, and let $P_{\mathcal{N}_t} = Q_t Q_t^\top$ be the Euclidean projector onto the current neutral subspace. Given the previously held portfolio u_{t-1}, define its current neutral projection by*

$$u_{t-1\to t} := P_{\mathcal{N}_t} u_{t-1}.$$

Let $u_t^* \in \mathcal{N}_t$ be the full neutral target, and set

$$d_t := u_t^* - u_{t-1\to t}.$$

For $\beta \in [0,1]$, define

$$u_t(\beta) := u_{t-1\to t} + \beta d_t.$$

Then:

1. *For every $\beta \in [0,1]$, $F_t^\top u_t(\beta) = 0$.*
2. *The distance from the projected previous holding to the full neutral target satisfies*
$$\|u_t(\beta) - u_{t-1\to t}\|_1 = \beta \|u_t^* - u_{t-1\to t}\|_1. \tag{26}$$
3. *The actual trade from the previously held portfolio satisfies*
$$\|u_t(\beta) - u_{t-1}\|_1 \leq \|u_{t-1\to t} - u_{t-1}\|_1 + \beta \|u_t^* - u_{t-1\to t}\|_1. \tag{27}$$
4. *Suppose $\Sigma_t \succeq 0$, $\gamma > 0$, $c \geq 0$, and $d_t^\top \Sigma_t d_t > 0$. Then the one-dimensional turnover-penalised objective*
$$\Phi_t(\beta) = a_t^\top u_t(\beta) - \frac{\gamma}{2} u_t(\beta)^\top \Sigma_t u_t(\beta) - c\|u_t(\beta) - u_{t-1}\|_1 \tag{28}$$
is concave and piecewise quadratic on $[0,1]$. Its global maximiser is obtained by checking the endpoints, all sign-change breakpoints
$$\mathcal{B}_t := \{\beta \in (0,1) : u_{t-1\to t,i} + \beta d_{t,i} - u_{t-1,i} = 0 \text{ for some } i \text{ with } d_{t,i} \neq 0\},$$
and the interval-wise critical points between consecutive breakpoints.

Proof. First, $u_{t-1\to t} = P_{\mathcal{N}_t} u_{t-1} \in \mathcal{N}_t$ by definition of the orthogonal projector. Also $u_t^* \in \mathcal{N}_t$ by assumption. Hence

$$d_t = u_t^* - u_{t-1\to t} \in \mathcal{N}_t,$$

because $\mathcal{N}_t$ is a linear subspace. Therefore $u_t(\beta) = u_{t-1\to t} + \beta d_t \in \mathcal{N}_t$ for every $\beta \in [0,1]$. Since $\mathcal{N}_t = \ker F_t^\top$, this proves $F_t^\top u_t(\beta) = 0$.

Second,

$$u_t(\beta) - u_{t-1\to t} = \beta(u_t^* - u_{t-1\to t}).$$

By positive homogeneity of the ℓ_1-norm and $\beta \geq 0$, this gives the displayed equality.

Third,

$$u_t(\beta) - u_{t-1} = (u_{t-1\to t} - u_{t-1}) + \beta(u_t^* - u_{t-1\to t}).$$

The triangle inequality gives the actual-trade bound.

It remains to prove the line-search statement. Write $u_0 = u_{t-1\to t}$ and $d = d_t$. The Markowitz part of the objective is

$$a_t^\top (u_0 + \beta d) - \frac{\gamma}{2}(u_0 + \beta d)^\top \Sigma_t (u_0 + \beta d).$$

Expanding in β, this equals

$$\text{constant} + \beta\left(a_t^\top d - \gamma d^\top \Sigma_t u_0\right) - \frac{\gamma}{2}\beta^2 d^\top \Sigma_t d.$$

Since $d^\top \Sigma_t d > 0$, this is strictly concave quadratic in β.

The turnover term is

$$\|u_0 + \beta d - u_{t-1}\|_1 = \sum_i |u_{0,i} + \beta d_i - u_{t-1,i}|.$$

Each summand is convex and piecewise linear in β, with a possible kink only at $u_{0,i} + \beta d_i - u_{t-1,i} = 0$. Therefore the negative turnover penalty is concave and piecewise linear. Hence Φ_t is concave and piecewise quadratic.

On any open interval between consecutive breakpoints, the sign vector $s_i = \operatorname{sign}(u_{0,i} + \beta d_i - u_{t-1,i})$ is constant. On such an interval,

$$\|u_0 + \beta d - u_{t-1}\|_1 = s^\top(u_0 + \beta d - u_{t-1}),$$

so

$$\Phi_t'(\beta) = a_t^\top d - \gamma d^\top \Sigma_t u_0 - \gamma\beta d^\top \Sigma_t d - c\, s^\top d.$$

The unique critical point on that interval, if it lies in the interval, is therefore

$$\beta_s^* = \frac{a_t^\top d - \gamma d^\top \Sigma_t u_0 - c\, s^\top d}{\gamma d^\top \Sigma_t d}.$$

A concave piecewise quadratic function on a compact interval attains its maximum at either an endpoint, a breakpoint, or an interval-wise critical point. Checking this finite set therefore gives a global maximiser. □

Without this projection, a holding that was neutral at the previous rebalance can carry exposure under the current matrix. Projecting the previous holding into the current neutral subspace repairs this before partial trading is applied.

5.2 One-dimensional rebalancing coefficient

Projected partial rebalancing reduces the trading decision to a one-dimensional optimisation on a neutral line segment. Let

$$u_0 := u_{t-1\to t}, \qquad d := u_t^* - u_0, \qquad u_t(\beta) = u_0 + \beta d.$$

Both u_0 and u_t^* lie in $\mathcal{N}_t$, so every $u_t(\beta)$ is neutral. Ignoring the turnover penalty, define

$$\phi_0(\beta) = a_t^\top(u_0 + \beta d) - \frac{\gamma}{2}(u_0 + \beta d)^\top \Sigma_t (u_0 + \beta d). \tag{29}$$

Then

$$\phi_0(\beta) = \phi_0(0) + \beta\left(a_t^\top d - \gamma d^\top \Sigma_t u_0\right) - \frac{\gamma}{2}\beta^2 d^\top \Sigma_t d. \tag{30}$$

Proposition 7 (Closed-form no-cost partial-rebalance fraction)**.** *Assume $\Sigma_t \succeq 0$, $\gamma > 0$, and $d^\top \Sigma_t d > 0$. Without the turnover penalty, the optimal coefficient on $[0,1]$ is*

$$\beta_0^* = \Pi_{[0,1]}\left(\frac{a_t^\top d - \gamma d^\top \Sigma_t u_0}{\gamma d^\top \Sigma_t d}\right), \tag{31}$$

where $\Pi_{[0,1]}(x) = \min\{1, \max\{0, x\}\}$.

Proof. Differentiating the displayed quadratic gives

$$\phi_0'(\beta) = a_t^\top d - \gamma d^\top \Sigma_t u_0 - \gamma \beta d^\top \Sigma_t d.$$

Since $d^\top \Sigma_t d > 0$, the function is strictly concave in β. The unconstrained maximiser is the displayed fraction, and projecting this value onto $[0, 1]$ gives the constrained maximiser. □

Algorithm 1. Projected partial neutral-coordinate rebalancing

1. Construct the exposure matrix F_t from information available before the holding interval.
2. Compute an orthonormal basis Q_t of $\mathcal{N}_t = \ker F_t^\top$.
3. Compute the full neutral target $u_t^* = Q_t \theta_t^*$ by solving the neutral Markowitz problem in neutral coordinates.
4. Project the previous holding into the current neutral subspace:
$$u_{t-1\to t} = Q_t Q_t^\top u_{t-1}.$$
5. Set $d_t = u_t^* - u_{t-1\to t}$.
6. Choose $\beta_t \in [0, 1]$ by maximising
$$a_t^\top (u_{t-1\to t} + \beta d_t) - \frac{\gamma}{2}(u_{t-1\to t} + \beta d_t)^\top \Sigma_t (u_{t-1\to t} + \beta d_t) - c\|u_{t-1\to t} + \beta d_t - u_{t-1}\|_1.$$
7. Hold
$$u_t = (1 - \beta_t) u_{t-1\to t} + \beta_t u_t^*.$$
8. Record neutrality leakage $\|F_t^\top u_t\|_\infty$, turnover $\|u_t - u_{t-1}\|_1$, ex-ante risk, selected β_t, and realised holding-period return.

Table 1: Algorithmic summary of projected partial neutral-coordinate rebalancing.

5.3 Neutrality-preserving scalar overlays

Several implementation overlays multiply a portfolio by a scalar, for example volatility targeting, scalar leverage normalisation, or scalar risk-budget adjustment. Such overlays preserve the neutrality constraint.

Proposition 8 (Scalar overlays preserve capital-level neutrality)**.** *Let $u_t \in \ker F_t^\top$, and let $s_t \in \mathbb{R}$ be any scalar that may depend on lagged data, ex-ante volatility, leverage rules, or risk-budgeting rules. Then*

$$F_t^\top (s_t u_t) = 0.$$

Consequently, scalar volatility targeting, scalar gross-exposure normalisation, and scalar risk-budget adjustment do not introduce exposure leakage. More generally, if $u_t(\beta) \in \ker F_t^\top$ for every $\beta \in [0, 1]$, then

$$s_t(\beta) u_t(\beta) \in \ker F_t^\top$$

for every scalar function $s_t(\beta)$.

Proof. By linearity,

$$F_t^\top(s_t u_t) = s_t F_t^\top u_t = 0.$$

The same argument applies pointwise to $s_t(\beta)u_t(\beta)$. $\square$

Remark 2 (Non-scalar overlays can destroy neutrality)**.** *The scalar condition cannot be dropped. If an implementation overlay acts through a diagonal matrix*

$$D_t = \operatorname{diag}(d_{1,t}, \dots, d_{n_t,t})$$

rather than a scalar multiple of the identity, then a neutral portfolio $u_t \in \ker F_t^\top$ need not remain neutral after the overlay:

$$F_t^\top D_t u_t$$

is generally nonzero. Thus assetwise clipping, position-dependent shrinkage, and asset-specific leverage scaling can reintroduce exposure leakage unless they are followed by a projection back into $\ker F_t^\top$ or handled inside the constrained optimiser.

Lemma 1 (Projection repair after an implementation overlay)**.** *Let $v_t \in \mathbb{R}^{n_t}$ be any portfolio obtained after an arbitrary implementation operation, such as clipping, assetwise scaling, or external trade adjustment. Define*

$$\Pi_t v_t := Q_t Q_t^\top v_t.$$

Then $\Pi_t v_t \in \ker F_t^\top$. Moreover, among all neutral portfolios $u \in \ker F_t^\top$, $\Pi_t v_t$ is the unique Euclidean closest portfolio to v_t:

$$\Pi_t v_t = \operatorname*{arg\,min}_{u \in \ker F_t^\top} \|u - v_t\|_2.$$

Proof. Since $Q_t Q_t^\top$ is the orthogonal projector onto $\mathcal{N}_t = \ker F_t^\top$, its range is exactly $\mathcal{N}_t$. Therefore $\Pi_t v_t \in \mathcal{N}_t$, and hence $F_t^\top \Pi_t v_t = 0$.

The Euclidean projection theorem for closed linear subspaces gives that $Q_t Q_t^\top v_t$ is the unique point in $\mathcal{N}_t$ minimising the Euclidean distance to v_t. For completeness, write any $u \in \mathcal{N}_t$ as $u = \Pi_t v_t + w$, where $w \in \mathcal{N}_t$. Since $v_t - \Pi_t v_t \in \mathcal{N}_t^\perp$,

$$\|v_t - u\|_2^2 = \|v_t - \Pi_t v_t - w\|_2^2 = \|v_t - \Pi_t v_t\|_2^2 + \|w\|_2^2.$$

This is minimised uniquely when $w = 0$, i.e. when $u = \Pi_t v_t$. $\square$

6 Experimental design

6.1 Research universes and baselines

The experiment suite uses six research panels: US large-cap sample, Sector, Style, GlobalCountries, USBonds, and Country. The label "US large-cap sample" refers to a 48-asset cleaned level panel used in the local research files; it is not a historical S&P 500 membership file. The empirical comparison holds the alpha inputs and exposure construction fixed across construction rules. The core baselines are:

1. ordinary alpha, which uses the raw signal in the ordinary portfolio-construction layer;

2. residualised alpha, which first residualises the signal against F_t but does not enforce capital-space neutrality after construction;

3. full neutral-coordinate target, equivalently the full constrained neutral Markowitz target, which moves directly to the exact neutral target with no turnover-smoothing step;
4. projected partial neutral-coordinate portfolio, which projects the previously held portfolio into $\mathcal{N}_t$ and then partially rebalances according to (25).

The final fixed candidates are:

- Defensive momentum neutral-coordinate: a low-turnover candidate with β capped at 0.25;
- Ensemble continuous neutral-coordinate: the balanced empirical candidate using a signal ensemble and continuous β;
- Reversal continuous neutral-coordinate: a reversal-based comparison candidate with continuous β.

The candidates are fixed before the reported universe-level comparison and are not retuned separately by universe.

Universe	Sample period	Daily observations	Rebalances	Assets
Country	2006-02-28–2025-11-28	4,970	237	10
GlobalCountries	2006-01-04–2025-12-11	5,017	239	16
US large-cap sample	2011-01-04–2025-12-03	3,752	179	48
Sector	2001-01-03–2025-12-11	6,346	303	11
Style	2006-02-28–2025-11-28	4,970	237	10
USBonds	2006-01-04–2025-12-11	5,017	239	13

Table 2: Research-panel coverage used in the fixed validation protocol. Asset counts are the assets with non-empty weight histories in the local validation files.

The main text keeps the empirical design compact. The implementation recipe and fixed-candidate protocol are recorded in Appendix A, Table 6.

6.2 Validation metrics

The experiments report three classes of quantities. The first is neutrality leakage, measured by $\|F_t^\top u_t\|_\infty$ at each rebalance and aggregated by the maximum observed value. The second is portfolio performance and implementation quality: Sharpe ratio, median Sharpe across universes, turnover reduction versus the full neutral-coordinate target, and maximum-drawdown improvement. The third covers stability checks: stationary-bootstrap comparisons (Politis and Romano, 1994), transaction-cost sensitivity, subperiod aggregates, and exposure-stress linearity.

Unless otherwise stated, Sharpe ratios are computed from the walk-forward portfolio return series using the same sampling convention across all strategies and universes. Turnover is measured at rebalance dates after gross normalisation. "TO red." denotes turnover reduction relative to the full neutral-coordinate target, and "MaxDD imp." denotes improvement in maximum drawdown relative to that same target, so positive values are favourable. Exposure leakage is reported as the maximum monitored value of $\|F_t^\top u_t\|_\infty$ over rebalances.

The empirical comparison is not designed as a test of alpha discovery. All strategies use the same alpha inputs, rebalance dates, risk estimates, and exposure definitions. The empirical questions are whether neutral coordinates eliminate capital-level exposure leakage after optimisation and rebalancing, whether projected partial rebalancing reduces turnover without leaving the neutral subspace, and whether the results are stable under transaction costs, conditioning, and exposure perturbations.

Across the reported order-sweep and exposure-stress checks, the construction maintains numerical-level neutrality and satisfies the leakage-bound tests. Appendix A reports the corresponding counts and stress summaries.

7 Empirical results

7.1 Aggregate fixed validation

Table 3 reports the aggregate comparison across the six research universes. Ordinary alpha and residualised alpha are included to separate signal-level exposure control from capital-level exposure control. The full neutral-coordinate portfolio is the constrained neutral target. The projected partial neutral-coordinate portfolios use the same neutral target but trade only part way toward it.

The empirical tests focus on exposure control and turnover reduction across research universes. The main questions are whether the construction method controls ex-post exposure leakage and whether projected partial rebalancing reduces turnover relative to the full neutral target while preserving the same monitored neutrality.

Strategy	Avg Sharpe	Median	Wins	TO red.	MaxDD imp.	Max exposure
Ordinary alpha	-0.189	-0.144	3	0.088	-0.103	14.312
Residualised alpha	-0.221	-0.194	0	-0.009	-0.017	2.785
Full neutral-coordinate target	-0.165	-0.144	0	0.000	0.000	4.83×10^{-15}
Defensive momentum neutral-coordinate	0.059	0.021	6	0.779	0.173	1.76×10^{-15}
Ensemble continuous neutral-coordinate	0.104	0.088	6	0.594	0.139	2.65×10^{-15}
Reversal continuous neutral-coordinate	0.194	0.193	5	0.230	0.124	4.97×10^{-15}

Table 3: Aggregate benchmark table across six research universes. Wins are Sharpe wins versus the full neutral-coordinate target. Ordinary and residualised alpha are included as diagnostic baselines: they illustrate that residualising the forecasting signal is not equivalent to imposing neutrality on the implemented capital vector. The main construction metrics are turnover reduction and maximum ex-post exposure leakage.

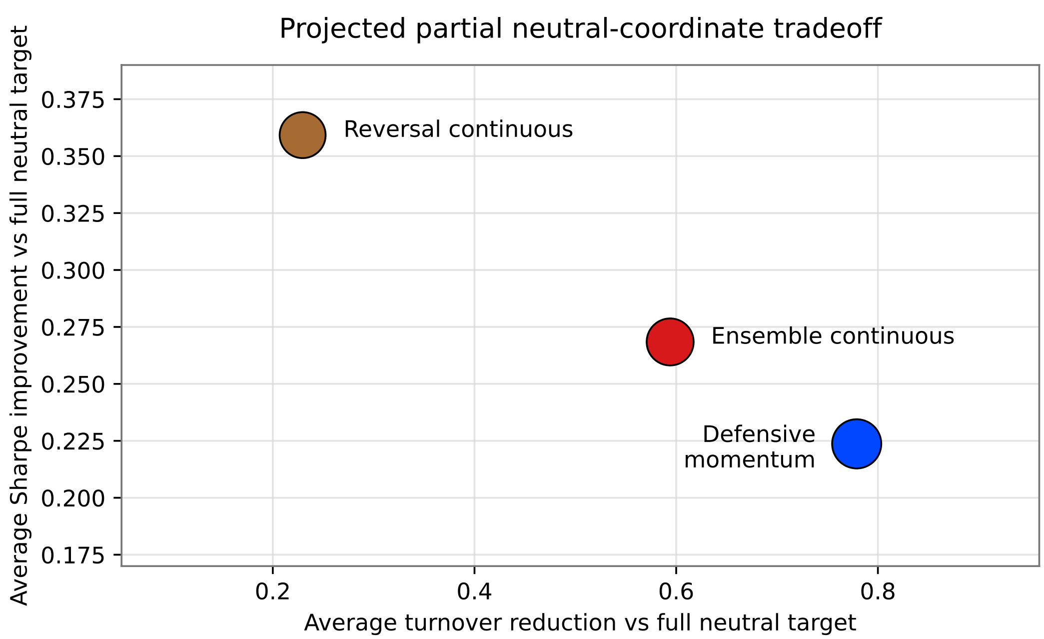

Figure 1: Projected partial neutral-coordinate candidates compared with the full neutral target. The horizontal axis reports average turnover reduction, and the vertical axis reports average Sharpe difference. Marker size is proportional to average maximum-drawdown improvement.

The Sharpe and drawdown columns are reported as implementation diagnostics. The primary empirical objects are exposure leakage, turnover, and the stability of the projected partial path after optimisation and rebalancing. The positive Sharpe differences in Table 3 indicate that the turnover-saving variants do not mechanically sacrifice realised performance in these panels, while the main construction result is the numerical enforcement of the monitored exposure constraints.

Per-universe rows are reported in Appendix A, Table 7. The appendix table records the heterogeneity of turnover reduction across panels.

The maximum exposure error among fixed neutral-coordinate candidates is 4.97×10^{-15}. For comparison, ordinary alpha has maximum exposure 14.312 and residualised alpha has maximum exposure 2.785 in the same final table. This provides the most direct empirical distinction between capital-space neutrality and signal residualisation in the reported tests.

7.2 Beta coefficients and neutral-coordinate conditioning

Table 4 reports the partial-rebalance coefficient summaries for the three fixed projected-partial candidates. The coefficients are not simply a refusal to trade: the defensive candidate has a binding cap by design, while the continuous candidates retain a substantial fraction of interior choices. Additional rank and conditioning checks for the neutral-coordinate basis are reported in Appendix A.

Strategy	Mean β	Median β	$\Pr(\beta = 0)$	$\Pr(\beta = 1)$	$\Pr(0 < \beta < 1)$	Turnover
Defensive momentum	0.175	0.250	0.038	0.000	0.962	0.196
Ensemble continuous	0.315	0.120	0.097	0.183	0.720	0.362
Reversal continuous	0.465	0.230	0.095	0.379	0.526	0.682

Table 4: Projected partial rebalancing coefficient summaries, averaged across the six universes in the fixed validation protocol.

7.3 Exposure specification and stress sensitivity

The main validation uses $F_t = [1, \sigma, \sigma^2, \sigma^3]$. The cubic volatility specification is used because it gives a compact test of neutrality against both linear and nonlinear characteristic exposure without introducing a large multi-factor risk model. An order-sweep check confirms that exact neutrality is not tied to the cubic exposure specification. The construction remains neutral when the exposure matrix is varied from $[1, \sigma]$ through $[1, \ldots, \sigma^4]$, with rank failures equal to zero in the reported sweep. The cubic specification is therefore used as the fixed validation choice rather than as a mathematically privileged exposure order. A stress test perturbs the exposure model and compares the realised leakage with the finite-sample misspecification bound.

Detailed stress-test summaries are reported in Appendix A, Table 8. The maximum bound ratio remains below one and the minimum linearity R^2 is 0.970, supporting the finite-sample misspecification bound in Proposition 6 without adding another main-text table.

7.4 Bootstrap uncertainty

Appendix A, Table 9, reports stationary-bootstrap uncertainty. Turnover reductions are stable across all six universes for the defensive and ensemble candidates; Sharpe and drawdown comparisons are favourable but less uniform. These results are consistent with the implementation-oriented scope of the study.

7.5 Transaction-cost sensitivity

Figure 2 reports sensitivity to realised transaction-cost assumptions. The comparison is relative to the full neutral target. As costs rise, the projected partial portfolios benefit from lower turnover.

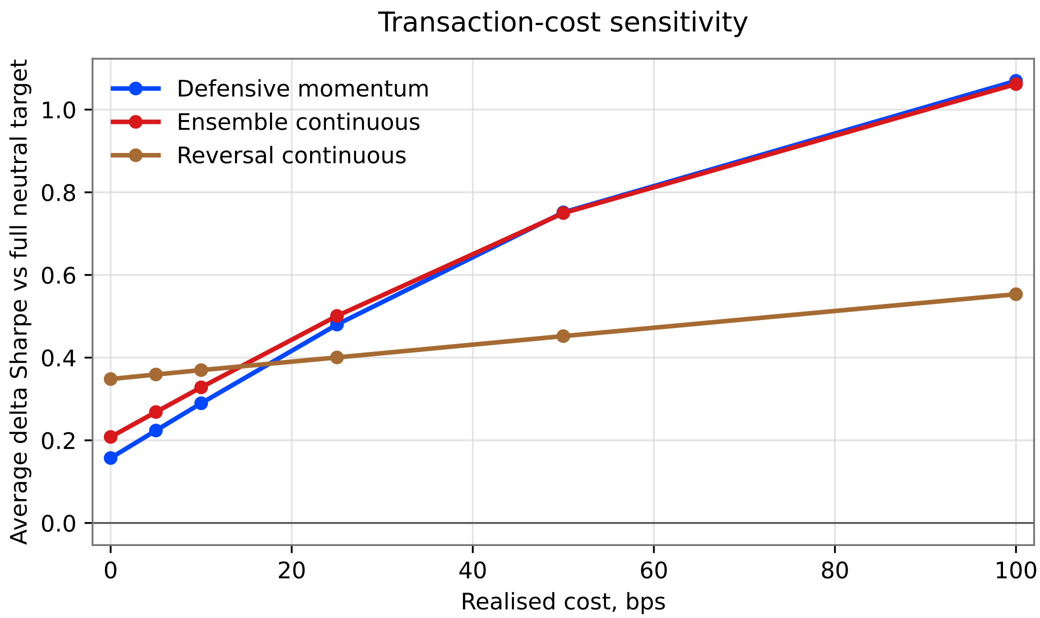

Figure 2: Transaction-cost sensitivity of the three fixed projected partial neutral-coordinate variants. The vertical axis reports average delta Sharpe versus the full neutral-coordinate target at the indicated realised-cost assumption. The figure reports a relative turnover-control diagnostic under simple realised-cost assumptions.

Early, middle, and late subperiod checks are directionally consistent with the aggregate interpretation. Turnover reductions persist across subperiods for the projected partial neutral-coordinate candidates, while Sharpe and drawdown improvements are favourable but less uniform. This supports the paper's main implementation claim without turning the subperiod split into a second model-selection exercise.

8 Research data audit and implementation diagnostics

The empirical panels are processed validation panels used to evaluate the portfolio-construction layer. The neutrality checks are computed directly from the implemented weights and the recorded exposure matrices at each rebalance. Country and Style have local OHLCV proxy coverage with volume and spread-proxy fields, while GlobalCountries has partial OHLCV proxy coverage for the available overlap. The US large-cap sample, Sector, and USBonds are cleaned level-only panels in the current data files. The US large-cap sample contains 48 series and is used as a fixed large-cap research sample.

The monitor records neutrality, selected beta, turnover, ex-ante risk, condition number, drawdown, monitoring flags, weights, and trade lists at every rebalance. The maximum monitored neutrality error is 4.94×10^{-15}; after scalar volatility targeting it is 6.00×10^{-15}, in line with Proposition 8. These values support the scalar-overlay invariance result without requiring a separate monitor table.

Proxy cost-accounting checks were computed only for panels where local OHLCV proxy fields are available, including partial coverage for GlobalCountries. These checks validate turnover and cost-

Universe	Source type	Volume/spread proxy	Diagnostics used
Country	local proxy OHLCV panel	yes	construction + cost-accounting check
GlobalCountries	local proxy OHLCV panel	partial	construction + partial cost-accounting check
US large-cap sample	cleaned level panel	no	construction and neutrality checks
Sector	cleaned level panel	no	construction and neutrality checks
Style	local proxy OHLCV panel	yes	construction + cost-accounting check
USBonds	cleaned level panel	no	construction and neutrality checks

Table 5: Research data-audit scope. The table records which checks are available in the replication files for the processed research panels.

accounting implementation; the detailed rows are included with the replication supplement rather than used as a main manuscript result.

9 Discussion

The empirical results support the finite-sample construction claim. Neutral-coordinate portfolios have exposure leakage at numerical precision, while ordinary and residualised-alpha portfolios retain material exposure after optimisation and rebalancing. This distinction is the main practical point of the paper: residualising a signal and enforcing neutrality on the traded capital vector are different operations.

The projected partial portfolios illustrate the role of the projection step. When the exposure matrix changes through time, the previous holding is not generally neutral under the new exposure matrix. Projecting the old holding into the current neutral subspace creates a neutral starting point for the new rebalance. The subsequent interpolation controls turnover while remaining inside the monitored neutral subspace.

The performance statistics provide implementation context. In the reported panels, projected partial neutral-coordinate portfolios reduce turnover relative to the full neutral target while preserving capital-level neutrality. The same construction layer can be combined with richer alpha models, commercial or internal risk models, and stricter liquidity and capacity controls. Its role is to enforce the selected exposure policy on the implemented portfolio after optimisation, scaling, and rebalancing.

10 Scope and extensions

The construction guarantees are finite-sample and exposure-relative. Given the monitored exposure matrix F_t, the neutral-coordinate portfolio lies in $\ker F_t^\top$, and the projected partial rebalancing path remains in the same current-date neutral subspace. The method therefore enforces the selected exposure policy on the implemented capital vector.

The empirical study uses fixed research panels, simple signal definitions, trailing covariance estimates, and a parsimonious volatility-exposure specification. These choices keep the validation focused on the construction layer. A production implementation can use a richer alpha model, a commercial or internal risk model, broader exposure definitions, liquidity constraints, and more detailed transaction-cost models without changing the neutral-coordinate construction.

The exposure matrix remains an investment-policy choice. The method enforces neutrality with respect to the columns included in F_t. Additional style, sector, country, beta, liquidity, or nonlinear characteristic exposures can be incorporated by adding columns to the monitored exposure matrix, subject to the available degrees of freedom in the universe.

11 Conclusion

This paper developed a finite-sample construction method for imposing factor neutrality on implemented portfolio weights. The key distinction is between residualising an alpha signal and constraining the capital vector that is actually traded. Signal residualisation can help interpret a forecast, but neutrality for an active sleeve is a property of the implemented weights after optimisation, scaling, and rebalancing.

At each rebalance, the exposure matrix F_t defines the monitored neutral capital subspace $\ker F_t^\top$. Writing the portfolio as $u_t = Q_t\theta_t$ restricts the optimiser to that subspace and gives the same capital vector as a Markowitz problem with equality constraints. The same representation yields finite-sample exposure-leakage bounds when the construction and monitoring exposure matrices differ.

The rebalancing rule projects the previous holding into the current neutral subspace before trading toward the new neutral target. A partial trade is then taken along the line segment between the projected holding and the full neutral target. Since both endpoints are neutral under the current exposure matrix, the entire interpolation path remains neutral.

In the reported walk-forward tests, neutral-coordinate portfolios have exposure leakage at numerical precision. The projected partial variants reduce turnover relative to the full neutral target while preserving capital-level neutrality. The construction provides a transparent way to implement and audit factor-neutral mandates at the level of traded capital weights.

Conflicts of Interest

The author declares no conflicts of interest.

Data and code availability

The empirical tables and figures are generated from research code implementing neutral-coordinate construction, projected partial rebalancing, and exposure-monitoring checks. Code sufficient to reproduce the reported tables and figures from the processed research panels will be provided as supplementary material or through an anonymised repository during review. Source data subject to vendor or licence restrictions are not redistributed.

A Additional technical and audit checks

This appendix records the projection ablation, fixed implementation protocol, per-universe rows, exposure-stress checks, stationary-bootstrap uncertainty, and rank-conditioning checks. Additional replication files include transaction-cost accounting details, subperiod summaries, exposure-order sweeps, monitor logs, and proxy cost-accounting rows.

A.1 Projection ablation: why partial rebalancing must start inside the current neutral subspace

The following ablation isolates the role of the projection step. Suppose the manager starts from the previous implemented portfolio u_{t-1} and partially rebalances directly toward the current full neutral target $u_t^* \in \mathcal{N}_t$, without first projecting u_{t-1} into the current neutral subspace. The resulting unprojected path is

$$\widetilde{u}_t(\beta) = (1-\beta)u_{t-1} + \beta u_t^*, \qquad 0 \le \beta \le 1. \tag{32}$$

Since $F_t^\top u_t^* = 0$, its current-date exposure is

$$F_t^\top \widetilde{u}_t(\beta) = (1-\beta) F_t^\top u_{t-1}. \tag{33}$$

Thus the unprojected path is current-date neutral only if $\beta = 1$ or the old holding already lies in the new neutral subspace $\mathcal{N}_t$. In walk-forward use, F_t changes with the exposure estimates, so yesterday's neutral portfolio need not be neutral under today's exposure matrix. The projected rule fixes this problem: with

$$u_{t-1\to t} = \Pi_{\mathcal{N}_t} u_{t-1}, \qquad u_t(\beta) = (1-\beta) u_{t-1\to t} + \beta u_t^*, \tag{34}$$

one has $F_t^\top u_t(\beta) = 0$ for every $\beta \in [0,1]$. The projection step is therefore required for exact capital-level neutrality under changing exposure definitions.

Component	Fixed validation specification
Rebalance frequency	Approximately monthly, implemented as 21-trading-day walk-forward rebalances.
Lookback window	252 trailing trading days, using only information available before the holding interval.
Exposure matrix	$F_t = [1, \sigma_t, \sigma_t^2, \sigma_t^3]$ in the fixed validation; $\sigma_{i,t}$ is the 252-day trailing volatility computed before the holding interval.
Risk estimate	Trailing covariance matrix with ridge regularisation in the optimisation step.
Signals	63-day momentum, 252-day momentum skipping 21 days, 21-day reversal, EWMA trend, volatility-scaled momentum, and equal-weight ensemble signals.
Fixed candidates	Defensive momentum, ensemble continuous, and reversal continuous neutral-coordinate.
Partial-rebalance rule	One-dimensional utility maximisation over $\beta \in [0,1]$.
Turnover penalty	Common fixed ex-ante coefficient 7500; it is held fixed across universes and candidates and was not chosen by universe-level performance tuning. The defensive candidate additionally caps β at 0.25.
Transaction-cost sensitivity	Reported at 0, 5, 10, 25, 50, and 100 basis points.
Universe-specific retuning	None in the fixed validation.

Table 6: Fixed implementation protocol. The table records the information set, candidate definitions, and common validation choices used in the final comparison.

Universe	Hard S	Mom S	Mom TO red.	Ens. S	Ens. TO red.	Rev. S
Country	-0.107	0.048	0.746	0.075	0.462	0.105
GlobalCountries	-0.490	-0.059	0.745	0.047	0.546	0.499
US large-cap	-0.181	-0.006	0.745	0.193	0.635	0.282
Sector	-0.237	0.117	0.804	0.159	0.661	0.321
Style	-0.067	-0.055	0.812	0.046	0.606	-0.060
USBonds	0.094	0.307	0.821	0.101	0.654	0.020

Table 7: Per-universe fixed neutral-coordinate comparison. "S" denotes Sharpe and "TO red." denotes turnover reduction versus the full neutral-coordinate target.

Universe	Max bound ratio	Median bound ratio	Min linearity R^2	Stress pass rate
Country	0.664	0.093	0.989	1.000
GlobalCountries	0.579	0.081	0.983	1.000
US large-cap	0.460	0.047	0.970	1.000
Sector	0.837	0.222	0.994	1.000
Style	0.763	0.115	0.989	1.000
USBonds	0.796	0.304	0.986	1.000

Table 8: Exposure-stress linearity and leakage-bound checks. The table summarises perturbed-exposure tests and supports the combined numerical and misspecification bound in Proposition 6.

Strategy	Metric	Avg delta	Win rate	CI excludes 0	N
Defensive momentum neutral-coordinate	Sharpe	0.224	0.869	2	6
Defensive momentum neutral-coordinate	Turnover	-0.689	1.000	6	6
Defensive momentum neutral-coordinate	Max drawdown	0.173	0.997	6	6
Ensemble continuous neutral-coordinate	Sharpe	0.268	0.861	3	6
Ensemble continuous neutral-coordinate	Turnover	-0.524	1.000	6	6
Ensemble continuous neutral-coordinate	Max drawdown	0.139	0.927	1	6
Reversal continuous neutral-coordinate	Sharpe	0.359	0.737	2	6
Reversal continuous neutral-coordinate	Turnover	-0.203	0.938	4	6
Reversal continuous neutral-coordinate	Max drawdown	0.124	0.812	1	6

Table 9: Stationary-bootstrap comparison against the full neutral-coordinate target. CI counts are the number of universes, out of six, where the 95 percent bootstrap interval excludes zero. Win rates are computed in the favourable direction: higher Sharpe, lower turnover, or improved maximum drawdown.

Universe	Median rank F_t	Min d_t	Median d_t	Median condition	Max leakage
Country	4	6	6	2.106	7.12×10^{-16}
GlobalCountries	4	12	12	3.563	1.54×10^{-15}
US large-cap	4	44	44	9.584	2.65×10^{-15}
Sector	4	7	7	1.028	4.81×10^{-16}
Style	4	6	6	2.032	6.48×10^{-16}
USBonds	4	9	9	6.019	9.07×10^{-16}

Table 10: Rank and conditioning checks for the main candidate monitor. Here $d_t = \dim \ker F_t^\top$ and the condition statistic is the monitored neutral-risk condition number for the neutral-coordinate risk problem.

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